

Theoretical Bounds on χ_ν in Observer-Patch Holography

Coherent-Matter Susceptibility and the Construction Window

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Abstract

OPH says that an ordinary tabletop gravity test on Earth should read null. The reason is simple. The canonical dark-sector response is controlled by

$$\rho_A = -\frac{1}{4\pi G} \nabla \cdot [(\nu_{\text{OPH}} - 1)\mathbf{g}_b],$$

and Earth sits deep in the high-acceleration regime where $\nu_{\text{OPH}} \simeq 1$.

The separate hypothesis is local. A coherent material state may add a small local correction

$$\delta\nu = \chi_\nu S_{\text{coh}}$$

to the same interpolation function. The constant is χ_ν . A bench experiment measures it or bounds it.

The useful part is the definition of S_{coh} . The scalar is built from record capacity, record stability, predictive boundary coupling, visible overlap coherence, stored coherent energy density, and substrate directness. It vanishes when the record and prediction conditions fail. It is also defined on the observer-facing quotient, so hidden implementation choices do not change it.

In the weak-field lab limit the force law is

$$F = \frac{g^2}{4\pi G} A_\perp \chi_\nu \Delta S_{\text{coh}}.$$

For areal load $\Sigma = M/A_\perp$, target response fraction f , and $\Delta S_{\text{coh}} = \Gamma_{\text{eff}} u/u_0$, the required value is

$$|\chi_\nu|_{\text{req}} = \frac{4\pi G \Sigma f}{g} \frac{u_0}{\Gamma_{\text{eff}} u}.$$

Write $\Sigma_2 = \Sigma/(10^2 \text{ kg m}^{-2})$. For $u_0 = 8.5 \times 10^{-10} \text{ J m}^{-3}$, $\Gamma_{\text{eff}} = 1$, and $u = 10^5\text{--}10^7 \text{ J m}^{-3}$, the calculated full-response threshold is

$$|\chi_\nu|_{\text{req}} \simeq f \Sigma_2 (7.3 \times 10^{-23} \text{ to } 7.3 \times 10^{-25}),$$

scaling as Γ_{eff}^{-1} . Macroscopic areal loads with $\Sigma_2 \sim 1\text{--}10$ sit in the $10^{-22}\text{--}10^{-24}$ band. The order-unity linear ceiling over the same stored-energy range is 8.5×10^{-15} to 8.5×10^{-17} , also scaled by Γ_{eff}^{-1} . For a declared response target, χ_ν has to lie between these two OPH-derived bounds. The result is a clean construction window for the continuation, separate from the recovered OPH core.

Contents

1	Status and Claim Boundary	2
2	Canonical Chain	3
2.1	Fixed-cutoff carrier	3
2.2	Mismatch and repair	3
2.3	Records and checkpoints	3
2.4	Gravity and dark response	4
3	The Coherence Scalar	4
4	Structural Properties	5
5	Weak-Field Response	5
6	The Susceptibility Window	6
7	Power and Stored-Energy Bounds	7
7.1	Driven coherent oscillations	7
7.2	Static elastic storage	8
8	Construction-Scale Consequences	8
9	Experimental Bound Formula	10
10	Compatibility with Existing Nulls	10
11	What OPH Alone Bounds	11
12	Open Normalizations	11
13	Conclusion	11

1 Status and Claim Boundary

The OPH core used here consists of finite patch carriers, mismatch-lowering repair, observer-accessible record algebras, quotient normal forms, checkpoint continuation, support-visible BW scaling, fixed-cap generalized-entropy stationarity, and the Jacobson-type Einstein branch. These ingredients are supplied by the consensus, microphysics, synthesis, and compact SM/GR papers [3,

4, 1, 2].

The dark-sector paper supplies the settled weak-field anomaly law

$$\rho_A = -\frac{1}{4\pi G} \nabla \cdot [(\nu_{\text{OPH}} - 1)\mathbf{g}_b], \quad (1.1)$$

interpreting the anomaly as a transported modular/collar information-defect remainder, with no new particle species postulated [6].

The χ_ν law is a continuation outside the recovered core:

$$\delta\nu(x, t) = \chi_\nu S_{\text{coh}}(x, t). \quad (1.2)$$

Here S_{coh} is defined from OPH-native observer-facing quantities, and the mathematical response window for χ_ν is derived. No claim is made that χ_ν is nonzero.

2 Canonical Chain

2.1 Fixed-cutoff carrier

At fixed cutoff, an observer-facing OPH implementation surface is a federated patch carrier

$$\mathfrak{F} = (V, E, \{\mathcal{A}_i\}_{i \in V}, \{\mathcal{I}_e\}_{e \in E}, \{\pi_{i,e}\}, \{\mathcal{R}_i\}, \{\mathcal{U}_i\}). \quad (2.1)$$

Here $\mathcal{A}_i$ are local finite algebras, $\mathcal{I}_e$ are overlap interfaces, $\pi_{i,e}$ are visible restriction maps, $\mathcal{R}_i \subseteq Z(\mathcal{A}_i)$ are record algebras, and $\mathcal{U}_i$ are local update and repair interfaces. Physical claims are made through visible restrictions, record algebras, and quotient-local observables, without choosing hidden microscopic representatives [4].

2.2 Mismatch and repair

The consensus paper gives a nonnegative visible mismatch functional Φ . Accepted repairs lower the relevant mismatch on finite state spaces. Under the declared local-diamond and repair-completeness hypotheses, the terminal observer-facing normal form is unique on the physical quotient and independent of asynchronous update schedule [3].

2.3 Records and checkpoints

For an observer-supporting subfederation U , the microphysics paper uses a checkpoint of the form

$$\text{Chk}_U(t) = (\mathcal{R}_U(t), \rho_U^{\text{acc}}(t), \mathfrak{I}_U^{\text{ext}}(t), \nu_{\geq t}, \mathfrak{B}_U(t)). \quad (2.2)$$

If two checkpoints agree on the accessible record algebra, accessible state, external interface, boundary-update schedule class, and provenance bundle, they induce the same continuation law on the observer-accessible event algebra. Approximate agreement gives a controlled total-variation bound [4].

2.4 Gravity and dark response

The compact SM/GR paper keeps the recovered core and downstream continuations separate. On the stated support-visible scaling branch, the OPH stack gives a Jacobson-type Einstein relation; dark-sector response laws are phenomenological continuations and do not belong to that recovered-core tier [2]. The dark paper then models the anomaly by Eq. (1.1).

On Earth, $g_b/a_0 \sim 10^{11}$, so the canonical ν_{OPH} term is pinned near unity. A terrestrial tabletop anomaly attributed only to the canonical OPH dark sector should read null. The χ_ν continuation is tested against that null baseline as a separate effect.

3 The Coherence Scalar

Definition 3.1 (Patch coherence factors). *Let U be an observer-supporting patch subfederation at cycle t , with observer-accessible record algebra $\mathcal{Z}_U^{\text{rec}}(t)$. Let $Y_U(t)$ be the induced record outcome variable, $X_{\partial U, t+1:t+h}$ the boundary packet over horizon h , and $\Phi_U(t)$ the visible mismatch on the relevant support. Define*

$$R_U(t; h) = [1 - d_{\text{TV}}(\text{Law}(Y_U(t+h)), \text{Law}(Y_U(t)))] e^{-\delta_{\text{rec}}(t)}, \quad (3.1)$$

$$P_U(t; h) = \frac{I(Y_U(t); X_{\partial U, t+1:t+h})}{H(X_{\partial U, t+1:t+h}) + \epsilon}, \quad (3.2)$$

$$C_U(t) = \left[1 - \frac{\Phi_U(t)}{\Phi_U^{\text{max}} + \epsilon} \right]_+. \quad (3.3)$$

Fix a reference capacity $\text{Cap}_0 > 0$ and set

$$\widehat{\text{Cap}}_U(t) = \frac{\text{Cap}(\mathcal{Z}_U^{\text{rec}}(t))}{\text{Cap}_0}. \quad (3.4)$$

The minimal patch-coherence factor is

$$\mathcal{K}_U(t; h) = \mathbf{1}_{\text{self-read}} \widehat{\text{Cap}}_U(t) R_U(t; h) P_U(t; h) C_U(t). \quad (3.5)$$

Definition 3.2 (OPH-native coherent-matter scalar). *Let $u_{\text{stored}}(U, t)$ be the stored coherent matter energy density in the substrate mode supported by U , let u_0 be a fixed reference energy density, and let $\varepsilon_{\text{sub}}(U) \in [0, 1]$ be the substrate directness factor. Define*

$$S_{\text{coh}}(U, t; h) = \varepsilon_{\text{sub}}(U) \frac{u_{\text{stored}}(U, t)}{u_0} \mathcal{K}_U(t; h). \quad (3.6)$$

Given a partition of unity $w_U(x)$ over observer-supporting subfederations, define the coarse-grained field

$$S_{\text{coh}}(x, t) = \sum_U w_U(x) S_{\text{coh}}(U, t; h), \quad \sum_U w_U(x) = 1. \quad (3.7)$$

The continuation axiom is the single statement

$$\boxed{\nu_{\text{eff}}(x, t) = \nu_{\text{OPH}}(x, t) + \chi_\nu S_{\text{coh}}(x, t).} \quad (3.8)$$

4 Structural Properties

Proposition 4.1 (Observer-facing quotient well-definedness). *Assume two microscopic representations induce the same observer-accessible continuation law on U , preserve the same visible mismatch score, and expose the same stored coherent energy density and substrate class. Then they give the same $S_{\text{coh}}(U, t; h)$.*

Proof. The factors $\text{Cap}(\mathcal{Z}_U^{\text{rec}})$, R_U , P_U , and C_U are defined from observer-accessible records, boundary-law variables, and visible mismatch data. These are precisely the quantities that survive the observer-facing quotient. Multiplying them by u_{stored} and ε_{sub} , which are declared observable substrate data for the continuation, does not introduce hidden representative dependence. $\square$

Proposition 4.2 (Null conditions). *If U has no self-read, no stable record algebra, or no predictive boundary coupling at horizon h , then $S_{\text{coh}}(U, t; h) = 0$.*

Proof. If there is no self-read, then $\mathbf{1}_{\text{self-read}} = 0$. If no stable record algebra is available, then either $\text{Cap}(\mathcal{Z}_U^{\text{rec}}) = 0$ or the stability factor R_U vanishes on the declared readout. If predictive boundary coupling is absent, then $P_U = 0$. In each case $\mathcal{K}_U = 0$, hence $S_{\text{coh}} = 0$. $\square$

Proposition 4.3 (Saturated-limit reduction). *If*

$$\mathbf{1}_{\text{self-read}} \simeq 1, \quad \widehat{\text{Cap}}_U \simeq 1, \quad R_U \simeq 1, \quad P_U \simeq 1, \quad C_U \simeq 1,$$

then

$$S_{\text{coh}}(U, t; h) \simeq \varepsilon_{\text{sub}}(U) \frac{u_{\text{stored}}(U, t)}{u_0}.$$

Proof. Substitute the saturated factor values into Eq. (3.6). $\square$

5 Weak-Field Response

Insert Eq. (3.8) into the settled weak-field density:

$$\rho_A^{\text{eff}} = -\frac{1}{4\pi G} \nabla \cdot [(\nu_{\text{OPH}} - 1 + \chi_\nu S_{\text{coh}}) \mathbf{g}_b]. \quad (5.1)$$

In a small terrestrial laboratory region with $\mathbf{g}_b \simeq -g \hat{z}$ and canonical $\nu_{\text{OPH}} - 1$ negligible, this becomes

$$\rho_A^{\text{lab}} \simeq -\frac{1}{4\pi G} \nabla \cdot [\chi_\nu S_{\text{coh}} \mathbf{g}]. \quad (5.2)$$

Theorem 5.1 (Planar response law). *Let a thin device have horizontal projected area $A_\perp$, and define the bottom-to-top coherence contrast*

$$\Delta S_{\text{coh}} = S_{\text{coh bottom}} - S_{\text{coh top}}.$$

Then, to leading order in the thin-device approximation, the apparent vertical force generated by the χ_ν continuation is

$$F_\chi = \frac{g^2}{4\pi G} A_\perp \chi_\nu \Delta S_{\text{coh}}. \quad (5.3)$$

Proof. With $\mathbf{g} = -g\hat{z}$, Eq. (5.2) gives

$$\rho_A^{\text{lab}} = \frac{g\chi_\nu}{4\pi G} \frac{\partial S_{\text{coh}}}{\partial z}$$

up to the sign convention for z . Integrating through a thin column gives

$$M_A = \int \rho_A^{\text{lab}} dV = -\frac{g\chi_\nu}{4\pi G} A_\perp \Delta S_{\text{coh}}$$

when ΔS_{coh} is bottom minus top. The apparent weight change is $F_\chi = -gM_A$, yielding Eq. (5.3). $\square$

Corollary 5.2 (Geometric ceiling). *If $|\Delta\nu| \lesssim 1$, then*

$$\frac{|F_\chi|}{A_\perp} \lesssim C_g \equiv \frac{g^2}{4\pi G} = 1.1466 \times 10^{11} \text{ N m}^{-2} \quad (5.4)$$

for $g = 9.80665 \text{ m s}^{-2}$ and $G = 6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$.

This ceiling is algebraic. It does not assert that any material system reaches $|\Delta\nu| = 1$.

6 The Susceptibility Window

For bounds it is useful to package the contrast geometry, substrate directness, and OPH coherence factors into one dimensionless number:

$$\Delta S_{\text{coh}} = \Gamma_{\text{eff}} \frac{u}{u_0}, \quad 0 \leq \Gamma_{\text{eff}} \leq 1. \quad (6.1)$$

Here u is the representative stored coherent energy density in the active mode. The factor Γ_{eff} includes vertical contrast, substrate directness, and effective $\mathcal{K}$.

Theorem 6.1 (Nonempty macroscopic linear-response window). *Let $\Sigma = M/A_\perp$ be the areal load, let $f \in [0, 1]$ be the target fraction of the ordinary weight Mg , and let $\epsilon_{\text{lin}} \leq 1$ be the maximum allowed $|\Delta\nu|$ in the chosen linear-response regime. Assume $\Gamma_{\text{eff}} > 0$. If*

$$f \frac{4\pi G \Sigma}{g} < \epsilon_{\text{lin}}, \quad (6.2)$$

then the interval

$$\boxed{\frac{4\pi G \Sigma f}{g} \frac{u_0}{\Gamma_{\text{eff}} u} \leq |\chi_\nu| \leq \epsilon_{\text{lin}} \frac{u_0}{\Gamma_{\text{eff}} u}} \quad (6.3)$$

is nonempty. Values in this interval produce at least fraction f of the ordinary weight inside the declared linear-response bound. Conversely, any value that produces at least fraction f while staying inside that linear-response bound lies in this interval.

Proof. The force law gives

$$\frac{|F_\chi|}{Mg} = \frac{g}{4\pi G \Sigma} |\chi_\nu| \Delta S_{\text{coh}}.$$

Requiring $|F_\chi|/(Mg) \geq f$ gives the lower bound in Eq. (6.3). Requiring $|\Delta\nu| = |\chi_\nu| \Delta S_{\text{coh}} \leq \epsilon_{\text{lin}}$ gives the upper bound. The interval is nonempty exactly when the lower bound is smaller than the upper bound, which is Eq. (6.2). $\square$

Corollary 6.2 (Numerical width in normalized areal load). *Let*

$$\Sigma_2 = \frac{\Sigma}{10^2 \text{ kg m}^{-2}}.$$

Then

$$\frac{4\pi G \Sigma}{g} = 8.5525 \times 10^{-9} \Sigma_2.$$

The full-response ($f = 1$), order-unity linear window has width

$$\frac{\chi_{\text{upper}}}{\chi_{\text{lower}}} = 1.1692 \times 10^8 \Sigma_2^{-1},$$

independent of u and Γ_{eff} . With $\epsilon_{\text{lin}} = 0.1$, the window width is $1.1692 \times 10^7 \Sigma_2^{-1}$.

$u [\text{J m}^{-3}]$	u/u_0	$ \chi_\nu _{\text{req}}$ for $f\Sigma_2/\Gamma_{\text{eff}} = 1$	$ \chi_\nu _{\text{lin}}$ for $\epsilon_{\text{lin}}/\Gamma_{\text{eff}} = 1$
1.0×10^5	1.18×10^{14}	7.27×10^{-23}	8.5×10^{-15}
5.5×10^5	6.47×10^{14}	1.32×10^{-23}	1.55×10^{-15}
1.0×10^6	1.18×10^{15}	7.27×10^{-24}	8.5×10^{-16}
4.0×10^6	4.71×10^{15}	1.82×10^{-24}	2.13×10^{-16}
1.0×10^7	1.18×10^{16}	7.27×10^{-25}	8.5×10^{-17}

Table 1: Full-response susceptibility bounds for $u_0 = 8.5 \times 10^{-10} \text{ J m}^{-3}$ and $\Sigma_2 = \Sigma/(10^2 \text{ kg m}^{-2})$. The third column should be multiplied by $f\Sigma_2/\Gamma_{\text{eff}}$. The fourth column should be multiplied by $\epsilon_{\text{lin}}/\Gamma_{\text{eff}}$.

The table gives the bound in numerical form. It says that ordinary material stress and coherent phonon energy densities put the macroscopic response threshold in the $\chi_\nu \sim 10^{-22}$ – 10^{-24} range for macroscopic areal loads and Γ_{eff} near unity. Smaller Γ_{eff} shifts the required value upward by $1/\Gamma_{\text{eff}}$.

7 Power and Stored-Energy Bounds

The force law fixes stored energy. Input power depends on how the coherent state is maintained.

7.1 Driven coherent oscillations

For a driven oscillator with angular frequency ω , mode volume V_{mode} , quality factor Q , and stored density u ,

$$P_{\text{loss}} = \frac{u V_{\text{mode}} \omega}{Q}. \quad (7.1)$$

Combining Eq. (7.1) with the required density from the macroscopic response law gives

$$\boxed{P_{\text{loss}} = \frac{4\pi G \Sigma f u_0 V_{\text{mode}} \omega}{g Q \Gamma_{\text{eff}} |\chi_\nu|}}. \quad (7.2)$$

For $\Sigma_2 = 1$, $V_{\text{mode}} = 10^{-3} \text{ m}^3$, $\omega = 2\pi(4.0 \times 10^4) \text{ s}^{-1}$, $Q = 10^5$, $f = 1$, and $\Gamma_{\text{eff}} = 1$, this gives:

$ \chi_\nu $	P_{loss}
10^{-20}	1.8 W
10^{-21}	18 W
10^{-22}	183 W
10^{-23}	1.8 kW
10^{-24}	18 kW

The scaling is $P_{\text{loss}} \propto (Q\Gamma_{\text{eff}}|\chi_\nu|)^{-1}$.

7.2 Static elastic storage

For a static elastic-strain coherent variable, the stored density is

$$u_{\text{elastic}} = \frac{\sigma^2}{2E}, \quad (7.3)$$

where σ is stress and E is Young's modulus. In that interpretation there is no oscillator decay factor ω/Q . Maintaining the state is then limited by creep, relaxation, and mechanical failure. Its loss law differs from Eq. (7.1). The same susceptibility bounds apply, because they depend only on u , Γ_{eff} , and Σ .

8 Construction-Scale Consequences

The bounds are algebraic. They do not prove that any substrate realizes $\Gamma_{\text{eff}} \simeq 1$, that $\chi_\nu \neq 0$, or that a high-energy coherent state can be maintained safely. They specify the requirements for a macroscopic force device if the continuation is real.

A force device, in this narrow sense, is a bounded substrate whose active region creates a vertical coherence contrast ΔS_{coh} . The apparent weight fraction is

$$\frac{|F_\chi|}{Mg} = \frac{g}{4\pi G\Sigma} |\chi_\nu| \Gamma_{\text{eff}} \frac{u}{u_0}. \quad (8.1)$$

All construction choices enter through the product

$$|\chi_\nu| \Gamma_{\text{eff}} u. \quad (8.2)$$

This is why χ_ν is the governing number. Geometry fixes the coefficient $g/(4\pi G\Sigma)$. Better materials can raise u . Better mode design can raise Γ_{eff} . A smaller χ_ν must be paid for linearly by higher stored energy density, larger active volume, better quality factor, or a smaller target response fraction. If $\chi_\nu = 0$, no amount of construction produces the continuation force.

For the normalized areal load $\Sigma_2 = 1$, the full-response condition is

$$\Delta\nu_{\text{full}}(\Sigma_2 = 1) = \frac{4\pi G\Sigma}{g} = 8.5525 \times 10^{-9}. \quad (8.3)$$

$ \chi_\nu $	u_{req} at $f\Sigma_2/\Gamma_{\text{eff}} = 1$	P_{loss} at $f\Sigma_2/\Gamma_{\text{eff}} = 1, Q = 10^5$	Construction class	Interpretation
10^{-20}	$7.3 \times 10^2 \text{ J m}^{-3}$	1.8 W	small bench article	The force is easy to produce if the coherence scalar is real. Existing ambient-coherence nulls become the main consistency check.
10^{-21}	$7.3 \times 10^3 \text{ J m}^{-3}$	18 W	tabletop module	Accessible to modest driven resonators or static strain specimens. A direct balance or pendulum experiment should see it cleanly.
10^{-22}	$7.3 \times 10^4 \text{ J m}^{-3}$	183 W	high-Q insert / compact platform	This is the first useful construction band. Full response is compatible with ordinary high-Q crystal or metal inserts. Thermal management matters.
10^{-23}	$7.3 \times 10^5 \text{ J m}^{-3}$	1.8 kW	engineered crystal or metal module	Inside high-strength material energy densities. Continuous driven operation becomes hard. Static elastic storage becomes much more attractive.
10^{-24}	$7.3 \times 10^6 \text{ J m}^{-3}$	18 kW	industrial multi-module system	Near the upper end of practical stored-density claims. Driven operation needs large area, high Q , cryogenic or exceptional materials, or partial-lift use.
10^{-25}	$7.3 \times 10^7 \text{ J m}^{-3}$	183 kW	large array / static-only candidate	Inside the response window. Compact driven construction is implausible. Static prestress, very high Γ_{eff} , or large arrays are the plausible routes.
10^{-26}	$7.3 \times 10^8 \text{ J m}^{-3}$	1.8 MW	outside compact construction	The response inequalities are satisfied. Material and power requirements exceed the intended compact-device scale. This is effectively a null for compact devices.

Table 2: Construction regimes inside the response-window theorem for $\Sigma_2 = 1$, $f = 1$, $u_0 = 8.5 \times 10^{-10} \text{ J m}^{-3}$, $V_{\text{mode}} = 10^{-3} \text{ m}^3$, $\omega = 2\pi(4.0 \times 10^4) \text{ s}^{-1}$, and $Q = 10^5$. Every row has $\Delta\nu = 8.5525 \times 10^{-9}$, so construction difficulty changes while algebraic admissibility is the same. Multiply the listed material and power requirements by $f\Sigma_2/\Gamma_{\text{eff}}$.

This is far below even a conservative $\epsilon_{\text{lin}} = 0.1$ linear bound. The required $\Delta\nu$ lies inside the theorem window. Construction depends on the required $\Gamma_{\text{eff}}u$ product, mode volume, quality factor, and material stress.

Two consequences follow.

First, a measurement of χ_ν selects a construction class. If $|\chi_\nu| \gtrsim 10^{-22}$ and Γ_{eff} is reasonably large, compact demonstrators are in the ordinary resonator and high-strength-substrate regime. If $|\chi_\nu| \sim 10^{-23}$ – 10^{-24} , the design must move toward high-Q crystals, larger active area, static strain, or partial lift. If $|\chi_\nu| \lesssim 10^{-25}$, compact force-device construction stops being the right first target. The useful output is a bound on the continuation.

Second, the static-versus-driven distinction is decisive. The driven column pays $uV_{\text{mode}}\omega/Q$. Static elastic storage pays no oscillator maintenance cost. It must survive creep and fracture. The same χ_ν value can be impractical for a driven phonon design and practical for a static strain design. The static-stress experiment decides which maintenance law applies to the construction table.

9 Experimental Bound Formula

Let an experiment have force sensitivity F_{min} , projected area $A_\perp$, and known coherence contrast ΔS_{coh} . A null result gives

$$|\chi_\nu| \leq \frac{4\pi G F_{\text{min}}}{g^2 A_\perp |\Delta S_{\text{coh}}|}. \quad (9.1)$$

If the mode is a driven oscillator, with $\Delta S_{\text{coh}} = \Gamma_{\text{eff}} Q P_{\text{in}} / (u_0 \omega V_{\text{mode}})$, then

$$|\chi_\nu| \leq \frac{4\pi G F_{\text{min}} u_0 \omega V_{\text{mode}}}{g^2 A_\perp \Gamma_{\text{eff}} Q P_{\text{in}}}. \quad (9.2)$$

If the mode is static elastic storage, replace the driven density by u_{elastic} from Eq. (7.3).

10 Compatibility with Existing Nulls

Proposition 10.1 (No ordinary-matter anomaly from null coherence). *If ordinary non-engineered configurations have $\mathcal{K}_U = 0$ on the observer-facing scalar of Eq. (3.5), then they impose no direct bound on χ_ν , because $\delta\nu = \chi_\nu S_{\text{coh}} = 0$ for those configurations.*

Proof. This is immediate from Eq. (3.6). The continuation couples only to the OPH-native coherent scalar. Undifferentiated rest mass or heat is outside this ansatz. $\square$

If ordinary matter has a small nonzero ambient scalar S_{cohamb} , then existing gravitational null tests bound the product:

$$|\chi_\nu| S_{\text{cohamb}} \leq \delta\nu_{\text{test}}. \quad (10.1)$$

This is a real consistency condition. It becomes a theorem-grade number when the ambient scalar is specified. The continuation is falsifiable in two ways: by direct χ_ν searches through Eq. (9.1) and by ordinary-gravity null tests once a model for S_{cohamb} is fixed.

11 What OPH Alone Bounds

The OPH-only result is conditional. For a declared areal load Σ , target fraction f , stored coherent density u , effective coherence Γ_{eff} , and linear allowance ϵ_{lin} , any χ_ν that produces the target response while staying inside the linear branch has to satisfy Eq. (6.3). Below the lower edge, the response is smaller than the target. Above the upper edge, the declared linear-response branch stops applying.

Proposition 11.1 (No OPH-only point value). *The recovered OPH core, together with the continuation axiom $\delta\nu = \chi_\nu S_{\text{coh}}$, does not fix a nonzero value of χ_ν , its sign, or a narrower absolute numerical interval than Eq. (6.3).*

Proof. The continuation term is outside the recovered Einstein/dark-sector core. The choice $\chi_\nu = 0$ returns exactly to that core and satisfies the quotient, checkpoint, null-coherence, and terrestrial high-acceleration baseline statements. The OPH construction also defines the response through the product $\chi_\nu S_{\text{coh}}$. Without a further theorem fixing a global maximum for S_{coh} , a canonical material value of u , or a variational principle for the continuation coefficient, the core theorems constrain only products and conditional response windows. Finite patch capacity makes the scalar well-defined at fixed cutoff; it does not select the numerical coupling. $\square$

Further narrowing needs one extra input: a canonical normalization of the maximum ordinary coherent scalar, an ambient-coherence theorem for ordinary matter, or a direct experimental bound. Those inputs can sharpen Eq. (6.3). They are separate from the recovered OPH core.

12 Open Normalizations

Three normalizations are parameters of the continuation.

First, Cap_0 must be fixed canonically. Depending on the substrate and regime, Cap could be $\log \dim \mathcal{Z}^{\text{rec}}$, an effective rank, usable record entropy, or effective patch capacity. The formulas keep Cap_0 explicit.

Second, the partition weights $w_U(x)$ should be induced by a declared observer-supporting subcover. Only the coarse-grained contrast ΔS_{coh} enters the bounds, so the construction is an implementation choice.

Third, the exponents in Eq. (3.5) are all set to one. A calibrated variant may replace

$$\mathcal{K}_U = \mathbf{1}_{\text{self-read}} \widehat{\text{Cap}_U} R_U P_U C_U$$

by a renormalized exponent family. The linear product is the minimal non-arbitrary continuation.

13 Conclusion

The χ_ν continuation can be stated as one additional local response law:

$$\nu_{\text{eff}} = \nu_{\text{OPH}} + \chi_\nu S_{\text{coh}}.$$

The scalar S_{coh} can be built from OPH-native observer-facing quantities: record capacity, record stability, boundary predictivity, visible overlap coherence, and stored coherent matter energy density. It is null when the observer-facing coherence conditions fail and quotient-invariant when the observer-accessible continuation law is unchanged.

The response bounds are simple. For an areal load Σ , target fraction f , and contrast $\Delta S_{\text{coh}} = \Gamma_{\text{eff}} u / u_0$,

$$|\chi_\nu|_{\text{req}} = \frac{4\pi G \Sigma f}{g} \frac{u_0}{\Gamma_{\text{eff}} u}.$$

The corresponding linear headroom extends up to

$$|\chi_\nu|_{\text{lin}} \leq \epsilon_{\text{lin}} \frac{u_0}{\Gamma_{\text{eff}} u}.$$

For $\Sigma_2 = \Sigma / (10^2 \text{ kg m}^{-2})$, the interval width is $1.1692 \times 10^8 \epsilon_{\text{lin}} / (f \Sigma_2)$. Coherent stored energy densities $10^5\text{--}10^7 \text{ J m}^{-3}$ put macroscopic full-response thresholds near $10^{-22}\text{--}10^{-24}$, scaled by $f \Sigma_2 \Gamma_{\text{eff}}^{-1}$. This is a mathematical continuation window outside the recovered OPH predictions and outside experimental results. The recovered core permits $\chi_\nu = 0$; a nonzero value or a tighter interval requires the additional inputs listed above.

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